

Jie Liu

Dual Master's in AI | Data & ML | Python · SQL · AWS

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SUMMARY

Data & AI practitioner completing a dual master's in AI and Machine Learning & Big Data, with hands-on experience across the full data lifecycle — designing AWS data pipelines, building and serving production ML models, writing SQL across relational databases, and turning raw data into dashboards that drive commercial decisions. Comfortable shipping features end-to-end with AI-assisted development, and keen to bring this to the Technology stream of the 2027 Kmart Group Graduate Program.

SKILLS

Languages: Python, R, SQL

Cloud & data engineering: AWS (Glue, S3, RDS, Redshift, EMR, Athena, DynamoDB, CloudFormation, SageMaker), ETL/ELT, data pipelines

Data warehousing & databases: Snowflake, Google BigQuery, OLAP, RDBMS, conceptual & logical data warehouse design, data modelling

Big data: Apache Hadoop, Hive, Spark, HBase, Kafka

Machine learning: scikit-learn, TensorFlow, Keras, NumPy, Pandas; regression, K-means, SVM, decision tree, random forest, GBMs, GLMs, KNN, PCA, Naïve Bayes, CNN, RNN, GNN

Visualisation: Tableau, Zoho Analytics

Professional: communication, project management

EDUCATION

Master by Research, Artificial Intelligence — UNSW Sydney

Sep 2024 – Dec 2026 (expected)

- Research focus: applications of Large Language Models (LLMs) in financial services.

Master of Computer Science, Machine Learning and Big Data (Coursework) —

Mar 2021 – Nov 2022

University of Wollongong

- Specialised in Machine Learning and Big Data; Distinction average (WAM: 81.3).
- High Distinctions in Programming & Data Structures, Data Management Systems, and Web Development; Capstone ranked Top 1 in cohort, Big Data Analytics Top 3.

PROFESSIONAL EXPERIENCE

AI Engineer Intern — Jobpin AI

Jan 2026 – Apr 2026

- Built an end-to-end production ML pipeline for personalised product recommendations, covering data ingestion, storage, transformation (feature engineering), and API serving on AWS.
- Contributed to the pipeline's production engineering alongside the team, including infrastructure-as-code, CI/CD, and security practices.
- Designed and implemented data pipelines on AWS to aggregate data from three external platforms, processing ~2,400 records/day.
- Standardised and transformed data formats to ensure accurate, consistent presentation across the company website.
- Developed automated data-expiry processes using two complementary approaches — a manual frontend trigger and a scheduled quarterly cleanup — improving data quality and system efficiency.
- Built backend and frontend features including an admin system and a user management page, using AI-assisted development to accelerate delivery.

Data Analyst — Hubify Limited (ASX: HFY)

Feb 2023 – Aug 2024

- Cleaned and validated 1M+ customer records through Python and SQL ETL pipelines, identifying and rectifying inaccuracies across the dataset.
- Built a recurring Python process to flag client records with stale details by cross-checking against external sources, flagging ~150 records/week and cutting manual verification by ~4 hrs/week.
- Designed and automated 6 monthly Tableau and Zoho Analytics dashboards, saving ~8 hrs/month and surfacing insights to support leadership's commercial decisions.
- Developed Python scripts to extract data from 4 platforms via APIs for downstream analysis and reporting.
- Delivered customer, market, and competitor analysis with tailored commercial recommendations, including segmentation dashboards covering sales and profit summaries, geographic and customer-segment breakdowns, and temporal trends.

Research Assistant — University of Wollongong

Sep 2022 – Nov 2022

- Employed Python to retrieve data on ~8,000 videos from www.bilibili.com using APIs for data analysis.
- Designed and implemented MySQL database tables to efficiently store and manage the retrieved JSON data comprising more than 20 distinct attributes.
- Conducted metric analysis to formulate independent variables, moderators, and dependent variables, subsequently used for model training.
- Trained and evaluated a linear regression model on video metadata.

PROJECTS

AWS Recommender System (Jobpin AI)

Built an end-to-end recommendation pipeline on AWS — data ingestion, feature engineering, Python ETL, model training, and ranked product recommendations — producing personalised rankings (CTR / click-intent) packaged for downstream use, with production-grade considerations such as data-quality checks and reproducible pipelines.

OzFish.AI — Android app to mitigate illegal fishing in NSW (Master's Capstone) ·

github.com/Jie1466803079/OzFish.AI

Led a six-person team to design, build, and test an Android app using transfer-learning models to identify 68 NSW fish species (90% top-5 accuracy) and an AR tool to measure catch length against NSW size regulations; ranked Top 1 in the cohort.

SCHOLARSHIPS

- **Postgraduate Academic Excellence Scholarship** — University of Wollongong (Master of Computer Science)
- **Research Training Program (RTP) Scholarship** — UNSW Sydney (Master by Research)
- **University Postgraduate Award (UPA)** — UNSW Sydney (Master by Research)

CERTIFICATIONS

- **Cybersecurity** — Hubify Limited (ASX: HFY)
- **NSW Data Analytics Virtual Experience Program**

LEADERSHIP & VOLUNTEERING

Volunteer (Data Science) — DataKind

Feb 2024 – Present

- **Technical Consultant, Climate Hack 2024** (Nov 2024 – Jan 2025) — Mentored teams of young changemakers from across Asia in this Singapore International Foundation programme building digital solutions to climate change, holding 1-on-1 consultations to advise on data science and AI, assess technical feasibility, and refine prototypes.
- **Proposal Reviewer, data.org AI2AI Challenge** (Jul – Aug 2024) — Scored first-round proposals for this global AI-for-social-impact and economic-inclusion initiative, backed by the Mastercard Center for Inclusive Growth, applying a structured rubric and giving constructive feedback to inform which applications advanced.

Marketing Manager — UOWCSSA (University of Wollongong student association)

Feb 2022 – Nov 2022

- Led a 4-person marketing team, managing the society's social media and designing promotional posters for 5 events.
- Organised two welcome nights for new students.